Namespace TUIS

Classes

Terminal

Class Terminal

Namespace: TUIS

Assembly: TUIS.dll

```
public class Terminal
```

Inheritance

object ← Terminal

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Terminal()

```
public Terminal()
```

Fields

Components

```
public readonly List<Component> Components
```

Field Value

List<Component>

InputSystem

```
public readonly InputSystem InputSystem
```

Field Value

InputSystem

NeedRedraw

```
public bool NeedRedraw
```

Field Value

bool

TimeSystem

```
public readonly TimeSystem TimeSystem
```

Field Value

TimeSystem

Methods

Clear()

```
public void Clear()
```

Draw()

```
public void Draw()
```

Start(Action<Terminal>?)

```
public void Start(Action<Terminal>? onStart = null)
```

Parameters

`onStart` Action<Terminal>

Namespace TUIS.Components

Classes

Component

Class Component

Namespace: TUIS.Components

Assembly: TUIS.dll

```
public class Component
```

Inheritance

object ← Component

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Component(Terminal, int, int, int, int)

```
public Component(Terminal terminal, int width, int height, int posY, int posX)
```

Parameters

`terminal` Terminal

`width` int

`height` int

`posY` int

`posX` int

Fields

Masks

```
public readonly List<Mask> Masks
```

Field Value

List<Mask>

Terminal

```
public Terminal Terminal
```

Field Value

Terminal

Properties

Height

```
public int Height { get; set; }
```

Property Value

int

IsVisible

```
public bool IsVisible { get; set; }
```

Property Value

bool

NeedRedraw

```
public bool NeedRedraw { get; set; }
```

Property Value

bool

PosX

```
public int PosX { get; set; }
```

Property Value

int

PosY

```
public int PosY { get; set; }
```

Property Value

int

Width

```
public int Width { get; set; }
```

Property Value

int

Methods

Draw()

```
public void Draw()
```

Namespace TUIS.Components.Masks

Classes

BackgroundMask

Print a single char in the background (Warning Masks will be printed in order so this one will override any Mask before itself)

BoxMask

A Mask must be attached to a Component and adds a layer of drawings on top of it. Masks will be drawn in the same order as they have been added to the Masks list.

This Mask will draw a box on the border of the Component.

ClockMask

A Mask must be attached to a Component and adds a layer of drawings on top of it. Masks will be drawn in the same order as they have been added to the Masks list.

This Mask will draw a clock and updates it accordingly to the real time.

CustomMask

A Mask must be attached to a Component and adds a layer of drawings on top of it. Masks will be drawn in the same order as they have been added to the Masks list.

This Mask helps to create a Mask with unique behaviour without needing to create an entire new class.

ImageMask

InputMask

Mask

A Mask must be attached to a Component and adds a layer of drawings on top of it. Masks will be drawn in the same order as they have been added to the Masks list.

TextMask

Enums

BoxMask.Type

The Type of boxing

Mask.BackgroundColor

Get the Mask.BackgroundColor using the ANSI color code.

Mask.TextColor

Get the Mask.TextColor using th ANSI color code.

Mask.TextDecoration

Get the Mask.TextDecoration using the ANSI color code.

TextMask.HorizontalAlignmentEnum

TextMask.VerticalAlignmentEnum

Class BackgroundMask

Namespace: TUIS.Components.Masks

Assembly: TUIS.dll

Print a single char in the background (Warning Masks will be printed in order so this one will override any Mask before itself)

```
public class BackgroundMask : Mask
```

Inheritance

object ← Mask ← BackgroundMask

Inherited Members

Mask.Component , Mask.NeedRedraw , Mask.IsVisible , Mask.Color , Mask.Background , Mask.Decoration , Mask.Draw() , Mask.DrawChar(int, int, char?, Mask.TextColor?, Mask.BackgroundColor?, Mask.TextDecoration?) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

BackgroundMask(Component, char, bool, TextColor, BackgroundColor, TextDecoration)

```
public BackgroundMask(Component component, char backgroundChar = ' ', bool isVisible
= true, Mask.TextColor color = TextColor.White, Mask.BackgroundColor background =
BackgroundColor.None, Mask.TextDecoration decoration = TextDecoration.Default)
```

Parameters

`component` Component

`backgroundChar` char

`isVisible` bool

`color` Mask.TextColor

`background` Mask.BackgroundColor

`decoration` Mask.TextDecoration

Properties

BackgroundChar

Holds the character to be drawn as the background.

```
public char BackgroundChar { get; set; }
```

Property Value

char

Methods

Behaviour()

This method is what makes the mask unique, it will describe that to print and where using the Draw Char(int, int, char?, TextColor?, BackgroundColor?, TextDecoration?). The method will automatically be called when needed.

This Mask will draw the BackgroundChar on top of the intire Component.

```
protected override void Behaviour()
```

Class BoxMask

Namespace: TUIS.Components.Masks

Assembly: TUIS.dll

A Mask must be attached to a Component and adds a layer of drawings on top of it. Masks will be drawn in the same order as they have been added to the Masks list.

This Mask will draw a box on the border of the Component.

```
public class BoxMask : Mask
```

Inheritance

object ← Mask ← BoxMask

Inherited Members

Mask.Component , Mask.NeedRedraw , Mask.IsVisible , Mask.Color , Mask.Background , Mask.Decoration , Mask.Draw() , Mask.DrawChar(int, int, char?, Mask.TextColor?, Mask.BackgroundColor?, Mask.TextDecoration?) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

BoxMask(Component, Type, bool, TextColor, BackgroundColor, TextDecoration)

Add a bounding box on the component.

```
public BoxMask(Component component, BoxMask.Type type, bool isVisible = true,
Mask.TextColor color = TextColor.White, Mask.BackgroundColor background =
BackgroundColor.None, Mask.TextDecoration decoration = TextDecoration.Default)
```

Parameters

`component` Component

`type` BoxMask.Type

The BoxMask.Type of boxing.

`isVisible` bool

`color` Mask.TextColor

`background` Mask.BackgroundColor

`decoration` Mask.TextDecoration

Properties

BoxType

Holds the BoxMask.Type of boxing.

```
public BoxMask.Type BoxType { get; set; }
```

Property Value

BoxMask.Type

Methods

Behaviour()

This method is what makes the mask unique, it will describe that to print and where using the DrawChar(int, int, char?, TextColor?, BackgroundColor?, TextDecoration?). The method will automatically be called when needed.

This Mask will draw a box on the border of the Component.

```
protected override void Behaviour()
```

Enum BoxMask.Type

Namespace: TUIS.Components.Masks

Assembly: TUIS.dll

The Type of boxing

```
public enum BoxMask.Type
```

Fields

`Double = 1`

`None = 3`

`Simple = 0`

`Thick = 2`

Class ClockMask

Namespace: TUIS.Components.Masks

Assembly: TUIS.dll

A Mask must be attached to a Component and adds a layer of drawings on top of it. Masks will be drawn in the same order as they have been added to the Masks list.

This Mask will draw a clock and updates it accordingly to the real time.

```
public class ClockMask : Mask
```

Inheritance

object ← Mask ← ClockMask

Inherited Members

Mask.Component , Mask.NeedRedraw , Mask.IsVisible , Mask.Color , Mask.Background , Mask.Decoration , Mask.Draw() , Mask.DrawChar(int, int, char?, Mask.TextColor?, Mask.BackgroundColor?, Mask.TextDecoration?) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ClockMask(Component, bool, TextColor, BackgroundColor, TextDecoration)

Add a real time clock to the Component. Warning: The clock is 27 wide by 3 tall. Any Component smaller than that will cause the clock to overshoot.

```
public ClockMask(Component component, bool isVisible = true, Mask.TextColor color =
TextColor.White, Mask.BackgroundColor background = BackgroundColor.None,
Mask.TextDecoration decoration = TextDecoration.Default)
```

Parameters

`component` Component

The component which the mask is attached to.

`isVisible` bool

Represent the visibility of the mask (default = true).

`color` Mask.TextColor

The default color of the mask (a mask's Behaviour()) method can override the color (default = white).

`background` Mask.BackgroundColor

The default background color of the mask (a mask's Behaviour()) method can override the background color (default = None).

`decoration` Mask.TextDecoration

The default decoration of the mask (a mask's Behaviour()) method can override the decoration (default = Default).

Methods

Behaviour()

This method is what makes the mask unique, it will describe that to print and where using the Draw Char(int, int, char?, TextColor?, BackgroundColor?, TextDecoration?). The method will automatically be called when needed.

This Mask's Behaviour() is to display large numbers .

```
protected override void Behaviour()
```

Class CustomMask

Namespace: TUIS.Components.Masks

Assembly: TUIS.dll

A Mask must be attached to a Component and adds a layer of drawings on top of it. Masks will be drawn in the same order as they have been added to the Masks list.

This Mask helps to create a Mask with unique behaviour without needing to create an entire new class.

```
public class CustomMask : Mask
```

Inheritance

object ← Mask ← CustomMask

Inherited Members

Mask.Component , Mask.NeedRedraw , Mask.IsVisible , Mask.Color , Mask.Background , Mask.Decoration , Mask.Draw() , Mask.DrawChar(int, int, char?, Mask.TextColor?, Mask.BackgroundColor?, Mask.TextDecoration?) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CustomMask(Component, Action<CustomMask>, bool, TextColor, BackgroundColor, TextDecoration)

Create a Mask with unique Behaviour() (`action`).

```
public CustomMask(Component component, Action<CustomMask> action, bool isVisible =
true, Mask.TextColor color = TextColor.White, Mask.BackgroundColor background =
BackgroundColor.None, Mask.TextDecoration decoration = TextDecoration.Default)
```

Parameters

`component` Component

`action` Action<CustomMask>

`isVisible` bool

`color` Mask.TextColor

`background` Mask.BackgroundColor

`decoration` Mask.TextDecoration

Fields

Properties

Holds properties used in the TUIS.Components.Masks.CustomMask._action. (Useless for now).

```
public readonly Dictionary<string, object> Properties
```

Field Value

Dictionary<string, object>

Methods

Behaviour()

This method is what makes the mask unique, it will describe that to print and where using the Draw Char(int, int, char?, TextColor?, BackgroundColor?, TextDecoration?). The method will automatically be called when needed.

```
protected override void Behaviour()
```

Class ImageMask

Namespace: TUIS.Components.Masks

Assembly: TUIS.dll

```
public class ImageMask : Mask
```

Inheritance

object ← Mask ← ImageMask

Inherited Members

Mask.Component , Mask.NeedRedraw , Mask.IsVisible , Mask.Color , Mask.Background , Mask.Decoration , Mask.Draw() , Mask.DrawChar(int, int, char?, Mask.TextColor?, Mask.BackgroundColor?, Mask.TextDecoration?) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

ImageMask(Component, string, bool, bool, TextColor, BackgroundColor, TextDecoration)

Draw an image in knida ASCII art, only windows compatible :,).

```
public ImageMask(Component component, string path, bool isColored = false, bool
isVisible = true, Mask.TextColor color = TextColor.White, Mask.BackgroundColor
background = BackgroundColor.None, Mask.TextDecoration decoration
= TextDecoration.Default)
```

Parameters

`component` Component

`path` string

`isColored` bool

`isVisible` bool

`color` Mask.TextColor

`background` Mask.BackgroundColor

`decoration` Mask.TextDecoration

Properties

IsColored

```
public bool IsColored { get; set; }
```

Property Value

bool

Methods

Behaviour()

This method is what makes the mask unique, it will describe that to print and where using the Draw Char(int, int, char?, TextColor?, BackgroundColor?, TextDecoration?). The method will automatically be called when needed.

```
protected override void Behaviour()
```

Export(string, int?, int?)

```
public void Export(string path = "./exported_image.png", int? exportedWidth = null,
int? exportedHeight = null)
```

Parameters

`path` string

`exportedWidth` int?

`exportedHeight` int?

Class InputMask

Namespace: TUIS.Components.Masks

Assembly: TUIS.dll

```
public class InputMask : Mask
```

Inheritance

object ← Mask ← InputMask

Inherited Members

Mask.Component , Mask.NeedRedraw , Mask.IsVisible , Mask.Color , Mask.Background , Mask.Decoration , Mask.Draw() , Mask.DrawChar(int, int, char?, Mask.TextColor?, Mask.BackgroundColor?, Mask.TextDecoration?) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

InputMask(Component, Action<InputMask>?, byte, byte, bool, TextColor, BackgroundColor, TextDecoration)

```
public InputMask(Component component, Action<InputMask>? onOutputChange = null, byte
horizontalPadding = 0, byte verticalPadding = 0, bool isVisible = true,
Mask.TextColor color = TextColor.White, Mask.BackgroundColor background =
BackgroundColor.None, Mask.TextDecoration decoration = TextDecoration.Default)
```

Parameters

`component` Component

`onOutputChange` Action<InputMask>

`horizontalPadding` byte

`verticalPadding` byte

`isVisible` bool

`color` Mask.TextColor

`background` Mask.BackgroundColor

`decoration` Mask.TextDecoration

Properties

Enabeled

```
public bool Enabeled { get; set; }
```

Property Value

bool

HorizontalPadding

```
public byte HorizontalPadding { get; set; }
```

Property Value

byte

Output

```
public string Output { get; }
```

Property Value

string

VerticalPadding

```
public byte VerticalPadding { get; set; }
```

Property Value

byte

Methods

Behaviour()

This method is what makes the mask unique, it will describe that to print and where using the Draw Char(int, int, char?, TextColor?, BackgroundColor?, TextDecoration?). The method will automatically be called when needed.

```
protected override void Behaviour()
```

Class Mask

Namespace: TUIS.Components.Masks

Assembly: TUIS.dll

A Mask must be attached to a Component and adds a layer of drawings on top of it. Masks will be drawn in the same order as they have been added to the Masks list.

```
public abstract class Mask
```

Inheritance

object ← Mask

Derived

BackgroundMask, BoxMask, ClockMask, CustomMask, ImageMask, InputMask, TextMask

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Mask(Component, bool, TextColor, BackgroundColor, TextDecoration)

A Mask goes in a Component and will add characters to print on it following custom rules: Behaviour().

```
public Mask(Component component, bool isVisible = true, Mask.TextColor color =
TextColor.White, Mask.BackgroundColor background = BackgroundColor.None,
Mask.TextDecoration decoration = TextDecoration.Default)
```

Parameters

`component` Component

The component which the mask is attached to.

`isVisible` bool

Represent the visibility of the mask (default = true).

`color` Mask.TextColor

The default color of the mask (a mask's Behaviour()) method can override the color (default = white).

`background` Mask.BackgroundColor

The default background color of the mask (a mask's Behaviour()) method can override the background color (default = None).

`decoration` Mask.TextDecoration

The default decoration of the mask (a mask's Behaviour()) method can override the decoration (default = Default).

Fields

Component

The Component the Mask is attached to.

```
public readonly Component Component
```

Field Value

Component

Properties

Background

Keeps the current Mask.BackgroundColor of the Mask.

```
public Mask.BackgroundColor Background { get; set; }
```

Property Value

Mask.BackgroundColor

Color

Keeps the current Mask.TextColor of the Mask.

```
public Mask.TextColor Color { get; set; }
```

Property Value

Mask.TextColor

Decoration

Keeps the current Mask.TextDecoration of the Mask.

```
public Mask.TextDecoration Decoration { get; set; }
```

Property Value

Mask.TextDecoration

IsVisible

Represent the visibility of the Mask.

```
public bool IsVisible { get; set; }
```

Property Value

bool

NeedRedraw

Helps to optimise the drawing of components, if NeedRedraw == false, the Mask won't be re-Draw()n. if NeedRedraw is set to true, it will automatically set the NeedRedraw to true.

```
public bool NeedRedraw { get; set; }
```

Property Value

bool

Methods

Behaviour()

This method is what makes the mask unique, it will describe that to print and where using the Draw Char(int, int, char?, TextColor?, BackgroundColor?, TextDecoration?). The method will automatically be called when needed.

```
protected abstract void Behaviour()
```

Draw()

This method will be called if the mask has changed and is visible. It calls the Behaviour() method of the mask.

```
public void Draw()
```

DrawChar(int, int, char?, TextColor?, BackgroundColor?, TextDecoration?)

Used in the Behaviour() method of Masks to properly print a character using relative coordinates.

```
protected void DrawChar(int y, int x, char? c, Mask.TextColor? textColor = null,
Mask.BackgroundColor? backgroundColor = null, Mask.TextDecoration? textDecoration
= null)
```

Parameters

y int

Coordinate on the y-axis (0 = top) relative to the Component.

x int

Coordinate on the x-axis (0 = left) relative to the Component.

`c` char?

The character to print (null = no print).

`textColor` Mask.TextColor?

The color which the `c` will be printed. null = the default background color of the mask.

`backgroundColor` Mask.BackgroundColor?

The background color which the `c` will be printed. null = the default background color of the mask.

`textDecoration` Mask.TextDecoration?

The decoration with which the `c` will be printed. null = the default decoration of the mask.

Enum Mask.BackgroundColor

Namespace: TUIS.Components.Masks

Assembly: TUIS.dll

Get the Mask.BackgroundColor using the ANSI color code.

```
public enum Mask.BackgroundColor
```

Fields

`Black = 40`

`Blue = 44`

`Cyan = 46`

`Green = 42`

`None = 0`

`Purple = 45`

`Red = 41`

`White = 47`

`Yellow = 43`

Enum Mask.TextColor

Namespace: TUIS.Components.Masks

Assembly: TUIS.dll

Get the Mask.TextColor using th ANSI color code.

```
public enum Mask.TextColor
```

Fields

`Black = 30`

`Blue = 34`

`Cyan = 36`

`Green = 32`

`Purple = 35`

`Red = 31`

`White = 37`

`Yellow = 33`

Enum Mask.TextDecoration

Namespace: TUIS.Components.Masks

Assembly: TUIS.dll

Get the Mask.TextDecoration using the ANSI color code.

```
public enum Mask.TextDecoration
```

Fields

`Bold = 1`

`Default = 0`

`Underline = 4`

Class TextMask

Namespace: TUIS.Components.Masks

Assembly: TUIS.dll

```
public class TextMask : Mask
```

Inheritance

object ← Mask ← TextMask

Inherited Members

Mask.Component , Mask.NeedRedraw , Mask.IsVisible , Mask.Color , Mask.Background , Mask.Decoration , Mask.Draw() , Mask.DrawChar(int, int, char?, Mask.TextColor?, Mask.BackgroundColor?, Mask.TextDecoration?) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

TextMask(Component, string, byte, byte, HorizontalAlignmentEnum, VerticalAlignmentEnum, bool, TextColor, BackgroundColor, TextDecoration)

```
public TextMask(Component component, string text, byte horizontalPadding = 0, byte
verticalPadding = 0, TextMask.HorizontalAlignmentEnum horizontalAlignment =
HorizontalAlignmentEnum.Left, TextMask.VerticalAlignmentEnum verticalAlignment =
VerticalAlignmentEnum.Top, bool isVisible = true, Mask.TextColor color =
TextColor.White, Mask.BackgroundColor background = BackgroundColor.None,
Mask.TextDecoration decoration = TextDecoration.Default)
```

Parameters

`component` Component

`text` string

`horizontalPadding` byte

`verticalPadding` byte

`horizontalAlignment` TextMask.HorizontalAlignmentEnum

`verticalAlignment` TextMask.VerticalAlignmentEnum

`isVisible` bool

`color` Mask.TextColor

`background` Mask.BackgroundColor

`decoration` Mask.TextDecoration

Properties

HorizontalAlignment

```
public TextMask.HorizontalAlignmentEnum HorizontalAlignment { get; set; }
```

Property Value

TextMask.HorizontalAlignmentEnum

HorizontalPadding

```
public byte HorizontalPadding { get; set; }
```

Property Value

byte

Text

```
public string Text { get; set; }
```

Property Value

string

VerticalAlignment

```
public TextMask.VerticalAlignmentEnum VerticalAlignment { get; set; }
```

Property Value

TextMask.VerticalAlignmentEnum

VerticalPadding

```
public byte VerticalPadding { get; set; }
```

Property Value

byte

Methods

Behaviour()

This method is what makes the mask unique, it will describe that to print and where using the Draw Char(int, int, char?, TextColor?, BackgroundColor?, TextDecoration?). The method will automatically be called when needed.

```
protected override void Behaviour()
```

Enum TextMask.HorizontalAlignmentEnum

Namespace: TUIS.Components.Masks

Assembly: TUIS.dll

```
public enum TextMask.HorizontalAlignmentEnum
```

Fields

`Center = 1`

`Left = 0`

`Right = 2`

Enum TextMask.VerticalAlignmentEnum

Namespace: TUIS.Components.Masks

Assembly: TUIS.dll

```
public enum TextMask.VerticalAlignmentEnum
```

Fields

`Bottom = 2`

`Center = 1`

`Top = 0`

Namespace TUIS.Systems

Classes

InputSystem

TimeSystem

Class InputSystem

Namespace: TUIS.Systems

Assembly: TUIS.dll

```
public class InputSystem : IDisposable
```

Inheritance

object ← InputSystem

Implements

IDisposable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Dispose()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources.

```
public void Dispose()
```

Start()

```
public void Start()
```

Events

OnKeyPress

```
public event Action<ConsoleKeyInfo>? OnKeyPress
```

Event Type

Action<ConsoleKeyInfo>

Class TimeSystem

Namespace: TUIS.Systems

Assembly: TUIS.dll

```
public class TimeSystem
```

Inheritance

object ← TimeSystem

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

TimeSystem()

```
public TimeSystem()
```

Fields

Timer

```
public readonly Timer Timer
```

Field Value

Timer

Properties

MiliSecond

```
public byte MiliSecond { get; }
```

Property Value

byte

Methods

AddTimedEvent(ElapsedEventHandler?)

```
public void AddTimedEvent(ElapsedEventHandler? eventHandler)
```

Parameters

`eventHandler` ElapsedEventHandler

Start()

```
public void Start()
```